CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

The first cohomology object in semi-abelian categories

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Abstract

We show that there is a natural notion of internal crossed homomorphism object of a split extension, defined as the representing object of a certain functor. Using these objects one can construct the first internal cohomology object of a split extension. We show that although these objects can be defined for an arbitrary split extension, they depend only on the center of the split extension. We show under suitable conditions that for abelian split extensions these internal cohomology objects coincide with those introduced in the authors PhD thesis.